

Question #1 of 120

Question ID: 1207263

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Which of the following statements about the CFA Institute's Professional Conduct Program (PCP) is *least accurate*?

- A) Possible sanctions include condemnation by a member's peers or suspension of a candidate's participation in the CFA Program.
- B) If the PCP staff determine that a sanction against a member is warranted, the member must either accept the sanction or lose the right to use the CFA designation.
- C) Members who cooperate with a PCP inquiry by providing confidential client information to PCP staff are not in violation of Standard III(E) Preservation of Confidentiality.

Explanation

Members can accept or reject a disciplinary sanction proposed by the Professional Conduct Program staff. If the member rejects the sanction, the matter is referred to a hearing before a disciplinary review panel of CFA Institute members. The other statements are accurate.

For Further Reference:

(Study Session 1, Module 2.1, LOS 2.a)

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Question #2 of 120

Question ID: 1207264

Robert Miguel, CFA, is a portfolio manager. On Saturday, one of his clients invited Miguel and his wife to be his guests at his luxury suite for a major league baseball playoff game, which they did. Miguel told his supervisor on Monday that they had attended the game with the client and that the suite was luxurious. Miguel has:

- A) not violated the Standards.
- B) violated the Standards because disclosure must be in writing.
- C) violated the Standards because he must disclose the gift prior to accepting.

Explanation

In this case, Miguel has not violated the standards. For a gift from a client in appreciation of past service or performance, informing his supervisor verbally is sufficient. Standard I(B) Independence and Objectivity requires disclosure prior to accepting the gift "when possible," but in cases such as this when there is short notice, notification afterward is permitted. (Module 3.1)

For Further Reference:

(Study Session 1, Module 3.1, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #3 of 120

Question ID: 1207272

At his golf club on Saturday morning, Paul Corwin, CFA, sees Frank Roberts, a friend and institutional client of his, who tells him that he is planning to sell his house on the 7th fairway. While golfing that day, Corwin tells Robert Lowe, a realtor, that Roberts is planning to sell his house and may need a realtor. He also tells Lowe that he manages an equities account for Roberts. If Corwin has not received permission from Roberts, he has violated the Standard on preservation of confidentiality:

- A) both by disclosing Roberts' plan to sell his home and that he is a client.
- B) by disclosing Roberts' plan to sell his home but not by mentioning that he was a client.
- C) by disclosing that Roberts is a client of his but not by mentioning Roberts' plan to sell his home.**

Explanation

Roberts violated Standard III(E) Preservation of Confidentiality by revealing his business relationship with Roberts without permission. Because the information that Roberts' plans to sell his home is not received as part of his professional relationship with Roberts, it is not covered by the Standard. (Module 3.5)

For Further Reference:

(Study Session 1, Module 3.5, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #4 of 120

Question ID: 1207266

Doug Watson, CFA, serves in a sales position at Sommerset Brokerage, a registered investment adviser. Watson frequently drinks excessively. On one occasion, Watson was cited by local police for misdemeanor public intoxication. According to the Standard on knowledge of the law and the Standard on misconduct, Watson is in violation of:

- A) both of these Standards.
- B) neither of these Standards.**
- C) only one of these Standards.

Explanation

Watson's excessive drinking is unfortunate, but we have no evidence that it has affected his work, professional integrity, judgment, or reputation. If he commits an act involving fraud or dishonesty, he would violate the Standard on misconduct. (Module 3.2)

For Further Reference:

(Study Session 1, Module 3.2, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #5 of 120

Question ID: 1207280

Reliable Wealth Managers wants to present a GIPS-compliant performance presentation and reference its GIPS compliance in marketing materials. Reliable is *least likely* required to:

- A) claim compliance with GIPS only if it has a compliant performance history of at least five years.**
- B) apply GIPS compliance firmwide and not only to the specific asset classes mentioned in the marketing materials.**
- C) include each discretionary fee-paying account in a composite based on its investment objectives and/or strategies.**

Explanation

GIPS require a firm to show a GIPS-compliant history for a minimum of five years, or since inception of the firm or the composite if in existence for less than five years. If Reliable has been in business for less than five years, it may still claim compliance with GIPS on a since-inception basis, provided the firm follows all other aspects of compliance. The other choices are requirements for a firm to claim compliance with GIPS.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.a)

CFA® Program Curriculum, Volume 1, page 235

Question #6 of 120

Question ID: 1207265

Peter Taylor, a CFA charterholder and a food industry analyst for a large investment firm, has been invited by Sweet Pineapple Co. to visit the firm's processing plants in Hawaii. The Standard concerning independence and objectivity recommends that Taylor:

- A) use and pay for commercial transportation, if available.**
- B) obtain written permission from his employer before he accepts this invitation.**
- C) decline this invitation if he issues recommendations on the firm's securities.**

Explanation

The recommended procedures for compliance with Standard I(B) Independence and Objectivity include the recommendation that analysts on company visits pay their own travel expenses and use commercial transportation if it is available. (Module 3.1)

For Further Reference:

(Study Session 1, Module 3.1, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #7 of 120

Question ID: 1207278

Ruth Brett, a Level I CFA candidate, feels nervous and unprepared the night before the exam. Brett writes a few key notes on the bottom of her shoe. At the exam, Brett sees the large number of proctors present and decides not to risk getting caught and does not look at her shoe. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, Brett is:

- A) not in violation of any Standard or the Code of Ethics because she did not use the notes.
- B) in violation of the Code of Ethics for bringing the notes into the examination room but is not in violation of any Standard because she did not use the notes.
- C) in violation of both the Code of Ethics and the Standard governing conduct as participants in CFA Institute programs for taking the notes into the examination room.**

Explanation

Brett violated both the Code of Ethics and Standard VII(A) Conduct as Participants in CFA Institute Programs. By writing down information from the Candidate Body of Knowledge and taking it into the exam room, she compromised the integrity of the exam, whether she used the notes or not. Her actions are also in violation of the Code of Ethics by not acting "with integrity, competence, diligence, respect, and in an ethical manner." (Module 3.9)

For Further Reference:

(Study Session 1, Module 3.9, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #8 of 120

Question ID: 1207267

Which of the following is *least likely* included in the CFA Code of Ethics? Members of CFA Institute must:

- A) place their clients' interests before their employer's interests.**
- B) strive to maintain and improve the competence of others in the profession.
- C) use reasonable care and exercise independent professional judgment.

Explanation

The requirement that members and candidates place their clients' interests before their employer's or their own is in Standard III(A) Loyalty, Prudence, and Care. The other choices are included in the CFA Institute Code of Ethics. (Module 3.4)

For Further Reference:

(Study Session 1, Module 3.4, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #9 of 120

Question ID: 1207275

In formulating her report on GammaCorp's common stock, Barb Kramer, CFA, did a complex series of statistical tests on the company's past sales and earnings. Based on this statistical study, Kramer stated in her report that, "GammaCorp's earnings growth for the next five years will average 15% per year." Her conclusion was based in part on a regression analysis with a high level of statistical significance. Has Kramer violated the Standard on communication with clients and prospective clients?

- A) Yes, because she didn't give complete details of the statistical model used.
- B) Yes, because she failed to indicate that 15% growth is an estimate.**
- C) No, because her projections are within the generally accepted bounds of statistical accuracy.

Explanation

Kramer violated Standard V(B) Communication with Clients and Prospective Clients. The problem is with the word "will." Kramer should have used "is estimated to be" to separate fact from opinion. Statistical estimates of future events are subject to change and should not be presented as certainties. She need not give complete details of the statistical model but should indicate its general characteristics and the important factors involved in her projections. (Module 3.7)

For Further Reference:

(Study Session 1, Module 3.7, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #10 of 120

Question ID: 1207268

Alpha Advisors, Inc., is an investment management firm with a client base that ranges from individuals to large foundations. Which of the following firm policies is *least appropriate* if Alpha adopts the Code and Standards? Alpha:

- A) monitors the personal trading activity of firm personnel and requires them to pre-clear personal trades.
- B) regularly calls larger accounts first after changes in investment recommendations have been faxed to all clients.
- C) excludes client accounts of family members of employees from participating in IPOs.**

Explanation

Standard III(B) Fair Dealing requires that all clients be treated fairly. Members and candidates should not discriminate against any client. A family member who is a fee-paying client should not be treated differently from other clients when taking investment action. Following up changes in recommendations with phone calls to larger clients is not a violation of the Standard if the changes have been disseminated fairly. (Module 3.4)

For Further Reference:

(Study Session 1, Module 3.4, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question ID: 1207269

Dudley Thompson is a bond salesman for a small broker/dealer in London. His firm is the lead underwriter on a new junk bond issue for Ibex Corporation, and Thompson has sent details of the offering to clients. Thompson calls only his accounts over £1,000,000 for whom he thinks the issue is suitable. Thompson also posts his firm's optimistic projections for Ibex's performance in several Internet chat rooms. According to the Standards concerning market manipulation and fair dealing, Thompson is in violation of:

- A) both of these Standards.
- B) **neither of these Standards.**
- C) only one of these Standards.

Explanation

Thompson has not violated Standard II(B) Market Manipulation by posting his firm's projections for Ibex. A firm's recommendation of a security may increase its price without any intent to mislead the market. The firm has disseminated the details of the offering to its clients fairly, so Thompson may call individual clients without violating the Standard III(B) Fair Dealing. (Modules 3.3 and 3.4)

For Further Reference:

(Study Session 1, Modules 3.3, 3.4, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #12 of 120

Question ID: 1207276

Rob Elliott, CFA, is an analyst with a large asset management firm. His personal portfolio includes a large amount of common stock of Tech Inc., a semiconductor company, which his firm does not currently follow. The director of the research department has asked Elliott to analyze Tech and write a report about its investment potential. Based on the CFA Institute Standards of Professional Conduct, the *most appropriate* course of action for Elliott is to:

- A) decline to write the report.
- B) sell his shares of Tech before completing the report.
- C) **disclose the ownership of the stock to his employer and in the report, if he writes it.**

Explanation

According to Standard VI(A) Disclosure of Conflicts, Elliott should disclose his beneficial ownership of Tech to his employer and to clients and prospects because such ownership could interfere with his ability to make unbiased and objective recommendations. Selling his shares or declining to write the report are not required and are more extreme than simply disclosing the potential conflict. (Module 3.8)

For Further Reference:

(Study Session 1, Module 3.8, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question ID: 1212077

Antonio Mendoza, CFA, an investment manager operating as AM Investments, solicits new business by making brief presentations at which he makes available a single-page information sheet that summarizes his performance history for the past 10 years. On the sheet, Mendoza has his phone number for those who would like more information along with the statement, "AM Investments has prepared and presented this report in compliance with the Global Investment Performance Standards (GIPS®)." Does Mendoza's brief presentation and information sheet violate GIPS?

- A) No.
- B) Yes, because the information sheet is inadequate.
- C) Yes, because the statement claims GIPS compliance for a single report.

Explanation

Mendoza has not violated the Standards of Practice or GIPS. Because the presentation was introductory and brief, Mendoza was not required to give any supporting documentation, but he made it available to clients and prospective clients upon request. His claim of GIPS compliance on the information sheet is appropriate.

For Further Reference:

(Study Session 1, Module 4.1, LOS 4.c)

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Question ID: 1207270

Anne Franklin, CFA, who covers technology stocks, joins a conference call for analysts presented by Cynthia Lucas, chief technology officer for LevelTech. Lucas tells the analysts that overseas shipments of the company's important new product are going to be delayed due to manufacturing defects, which is new information to the analysts. After the meeting Franklin changes her rating on LevelTech from "buy" to "hold" and sends a note to accounts recommending the sale of LevelTech. Franklin:

- A) did not violate the Standards.
- B) violated the Standard on nonpublic information by revising her rating on LevelTech.
- C) violated the Standard on fair dealing by rating the stock a "hold" but recommending sale of the shares to her accounts.

Explanation

Telling a selected group of analysts new information does not constitute public disclosure, and therefore acting or causing others to act on this information is a violation of Standard II(A) Material Nonpublic Information. Recommending the sale of a stock rated as a "hold" is not a violation of Standard III(B) Fair Dealing. (Modules 3.3 and 3.4)

For Further Reference:

(Study Session 1, Modules 3.3, 3.4, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #15 of 120

Question ID: 1207273

According to the recommended procedures for complying with the Standard on suitability, which of the following statements regarding an investment policy statement (IPS) is *least accurate*?

- A) An IPS should describe the roles and responsibilities of both the adviser and the client.
- B) A member or candidate is not responsible for financial information withheld by the client.
- C) **A client's IPS must be updated at least quarterly to reflect any changes in their investment profile.**

Explanation

Standard III(C) Suitability requires that members and candidates update client information (the IPS) at least annually. The IPS can be updated more frequently if there are significant changes in the investment strategy or client characteristics. (Module 3.5)

For Further Reference:

(Study Session 1, Module 3.5, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #16 of 120

Question ID: 1207274

Sue Johnson, CFA, has an elderly client with a very large asset base. The client intends to start divesting her fortune to various charities. Johnson is on the Board of a local charitable foundation. Johnson *most* appropriately:

- A) **must not discuss anything regarding her client and her client's intentions with the charitable foundation without permission.**
- B) can discuss her client's situation with the charitable foundation as long as she informs other local charities of her client's intentions.
- C) can make this known to the charitable foundation so that they can solicit the client, since it is the client's wish to divest assets to charities in the future.

Explanation

To comply with Standard III(E) Preservation of Confidentiality, Johnson must not discuss with her charitable foundation anything regarding her client and her client's intentions. It does not matter that her client intends to give money to charities in the near future. (Module 3.5)

For Further Reference:

(Study Session 1, Module 3.5, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #17 of 120

Question ID: 1207271

According to the Standard related to loyalty, prudence, and care, which of the following statements regarding the voting of proxies on client holdings is *least accurate*?

- A) Proxies have economic value to a client.
- B) An investment management firm should vote all proxies on client holdings unless the client reserves that right.**
- C) Members and candidates should explicitly disclose the firm's proxy voting policies to clients.

Explanation

Standard III(A) Loyalty, Prudence, and Care does not require the voting of all proxies. A cost-benefit analysis may support the conclusion that the voting of all proxies is not beneficial to the client in light of the time and effort required. Voting on nonroutine issues that have a material impact is required. (Module 3.4)

For Further Reference:

(Study Session 1, Module 3.4, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #18 of 120

Question ID: 1207277

Alvin Gold, CFA, resides in Country T and does business as an investment advisor primarily in Country U. Country T allows trading on non-public information and does not require disclosure of referral fees. Country U prohibits trading on non-public information only if it is gained by illegal means and requires disclosure of referral fees of over \$100 (U.S. equivalent). Gold accepts a referral fee of \$75, and in the course of a meeting with two other analysts and the firm's CFO, Gold receives material non-public information. To comply with the Code and Standards, Gold:

- A) need not disclose the referral fee but cannot trade on the non-public information.
- B) must disclose the referral fee and cannot trade on the non-public information.**
- C) must disclose the referral fee but may trade on the non-public information.

Explanation

According to Standard I(A) Knowledge of the Law, Gold must comply with the most strict of the laws of Country T, laws of Country U, and the CFA Standards of Practice. In this case, the most strict rules are those in the Standards of Practice. Standard VI(C) Referral Fees requires the disclosure of all referral fees and Standard II(A) Material Nonpublic Information prohibits acting or causing others to act on the basis of material non-public information. (Modules 3.1, 3.3, and 3.8)

For Further Reference:

(Study Session 1, Modules 3.1, 3.3, 3.8, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #19 of 120

Question ID: 1207281

Questions 19 through 30 relate to Quantitative Methods. (18 minutes)

An investor places \$5,000 in an account. The stated annual interest rate is 6% compounded monthly. The value of the account at the end of three years is *closest* to:

- A) \$5,970.
- B) \$5,978.
- C) **\$5,983.**

Explanation

With more than one compounding period per year, $FV_n = PV \left(1 + \frac{r_s}{m} \right)^{mn}$

$PV = \$5,000$; $r_s = 6\%$; $m = 12$; $\frac{r_s}{m} = \frac{6\%}{12} = 0.5\%$; $n = 3$; $m \times n = 12 \times 3 = 36$.

To compute FV_3 , enter $PV = -5,000$; $I = 0.5$; $PMT = 0$; and $N = 36$ to get \$5,983.40.

For Further Reference:

(Study Session 2, Module 6.1, LOS 6.d)

CFA® Program Curriculum, Volume 1, page 321

Question #20 of 120

Question ID: 1207289

Compared to a *t*-distribution with 10 degrees of freedom, and compared to a normal distribution, a *t*-distribution with 20 degrees of freedom and the same variance has:

Compared to df = 10

Compared to normal

- | | |
|------------------|---------------------|
| A) thinner tails | fatter tails |
| B) fatter tails | thinner tails |
| C) fatter tails | fatter tails |

Explanation

A *t*-distribution with sufficiently high degrees of freedom is approximately normal and a normal distribution has thinner tails compared to a *t*-distribution. The less the degrees of freedom, the fatter the tails.

For Further Reference:

(Study Session 3, Module 10.2, LOS 10.i)

CFA® Program Curriculum, Volume 1, page 584

Question #21 of 120

Question ID: 1207282

The initial market value of a portfolio was \$100,000. One year later the portfolio was valued at \$90,000 and two years later at \$99,000. The geometric mean annual return excluding any dividend income is *closest* to:

- A) -0.5%.
- B) -0.4%.
- C) 0.0%.

Explanation

$$R_1 = -10/100 = -10\%; R_2 = +9/90 = +10\%$$

$$\text{geometric mean} = \sqrt{(0.9)(1.1)} - 1 = -0.005 \text{ or } -0.5\%$$

An alternative way to get the geometric mean is:

$$\left(\frac{\text{ending value}}{\text{beginning value}} \right)^{\frac{1}{n}} - 1 = \text{geometric mean return}$$
$$\sqrt{\frac{99}{100}} - 1 = -0.005 \text{ or } -0.5\%$$

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.e)

CFA® Program Curriculum, Volume 1, page 386

Question #22 of 120

Question ID: 1207284

The probability that an acquisition has been executed well is 48%. If it is executed well, the probability of EPS greater than \$3.20 is 55%. If the acquisition has not been executed well, the probability of EPS less than or equal to \$3.20 is 65%. The unconditional probability of EPS greater than \$3.20 is *closest* to:

- A) 45%.
- B) 48%.
- C) 55%

Explanation

The unconditional probability, Prob(EPS > \$3.20), is equal to Prob(executed well) × Prob(EPS > \$3.20|executed well) + Prob(not executed well) × Prob(EPS > \$3.20|not executed well) = 0.48 × 0.55 + (1 – 0.48)(1 – 0.65) = 0.446.

For Further Reference:

(Study Session 2, Module 8.2, LOS 8.h)

Question #23 of 120

Question ID: 1207287

The "up-move factor" in a binomial tree is *best* described as:

- A) the probability that the variable increases in any period.
- B) one minus the "down-move factor" for the binomial tree.
- C) one plus the percentage change in the variable when it increases.**

Explanation

The up-move factor equals one plus the percentage increase when the variable goes up, and the down-move factor is equal to one divided by the up-move factor.

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.g)

CFA® Program Curriculum, Volume 1, page 526

Question #24 of 120

Question ID: 1207288

An analyst is asked to select a sample of securities from those included in a broad-based index that can be expected to have the same return as the index while preserving the key risk exposures of the index. The analyst should *most appropriately* use:

- A) simple random sampling.
- B) stratified random sampling.**
- C) constrained random sampling.

Explanation

Stratified random sampling is used to preserve characteristics of an underlying dataset.

For Further Reference:

(Study Session 3, Module 10.1, LOS 10.c)

CFA® Program Curriculum, Volume 1, page 516

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Question ID: 1207283

An analyst develops the following probability distribution for the states of the economy and market returns.

Unconditional Probability P(A)		Conditional Probability P(B A)	
Good economy	60%	Bull market	50%
		Normal market	30%
		Bear market	20%

Poor economy	40%	Bull market	20%
		Normal market	30%
		Bear market	50%

Which of the following statements about this probability distribution is *least accurate*?

- A) The unconditional probability of a normal market is 0.30.
- B) The joint probability of having a good economy and a bear market is 0.20.**
- C) Given that the economy is poor, the probability of a normal or a bull market is 0.50.

Explanation

$P(\text{good economy and bear market}) = 0.60 \times 0.20 = 0.12$. The other statements are true. The $P(\text{normal market}) = (0.60 \times 0.30) + (0.40 \times 0.30) = 0.30$. Given that the economy is poor, the probability of a normal or bull market $= 0.30 + 0.20 = 0.50$.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.f)

CFA® Program Curriculum, Volume 1, page 463

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Question ID: 1207285

An analyst estimates a stock has a 40% probability of earning a 10% return, a 40% probability of earning a 12.5% return, and a 20% probability of earning a 30% return. The stock's standard deviation of returns based on this returns model is *closest* to:

- A) 3.74%.
- B) 5.75%.
- C) 7.58%.**

Explanation

Expected value $= (0.4)(10\%) + (0.4)(12.5\%) + (0.2)(30\%) = 15\%$

Variance $= (0.4)(10 - 15)^2 + (0.4)(12.5 - 15)^2 + (0.2)(30 - 15)^2 = 57.5$

Standard deviation $= \sqrt{57.5} = 7.58\%$

For Further Reference:

(Study Session 2, Module 8.3, LOS 8.m)

CFA® Program Curriculum, Volume 1, page 489

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Question ID: 1207286

An investment manager has a pool of five security analysts he can choose from to cover three different industries. In how many different ways can the manager assign one analyst to each industry?

- A) 10.
- B) 60.**

C) 125.

Explanation

We can view this problem as the number of ways to choose three analysts from five analysts when the order they are chosen matters. The formula for the number of permutations is:

$$\frac{n!}{(n-r)!} = \frac{5!}{2!} = 5 \times 4 \times 3 = 60$$

On the TI financial calculator: $5 \text{ 2}^{\text{nd}} \text{ nPr } 3 = 60$.

Alternatively, there are $5 \text{ 2}^{\text{nd}} \text{ nCr } 3 = 10$ ways to select three of the five analysts, and for each group of selected analysts, there are $3! = 3 \times 2 \times 1 = 6$ ways to assign them the three industries. Therefore, there are $10 \times 6 = 60$ ways to assign the industries to the analysts.

For Further Reference:

(Study Session 2, Module 8.3, LOS 8.o)

CFA® Program Curriculum, Volume 1, page 495

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Question ID: 1207358

Shortfall risk is *best* described as the probability:

- A) of a credit rating downgrade due to possible earnings shortfalls.
- B) of failing to make a contractually promised payment.
- C) **that portfolio value will fall below some minimum level at a future date.**

Explanation

Choice A is downgrade risk; choice B is default risk.

For Further Reference:

(Study Session 3, Module 9.3, LOS 9.m, Study Session 15, Module 47.1, LOS 47.a)

CFA® Program Curriculum, Volume 1, page 537

CFA® Program Curriculum, Volume 5, page 302

Question #29 of 120

Question ID: 1207290

If a two-tailed hypothesis test has a 5% probability of rejecting the null hypothesis when the null is true, it is *most likely* that:

- A) the power of the test is 95%.
- B) **the confidence level of the test is 95%.**
- C) the probability of a Type I error is 2.5%.

Explanation

Rejecting the null hypothesis when it is true is a Type I error. The probability of a Type I error is the significance level of the test and one minus the significance level is the confidence level. The power of a test is one minus the probability of a Type II error, which cannot be calculated from the information given.

For Further Reference:

(Study Session 3, Module 11.1, LOS 11.c)

CFA® Program Curriculum, Volume 1, page 613

Question #30 of 120

Question ID: 1207291

Which of the following statements about hypothesis testing is *most accurate*?

- A) Rejecting a true null hypothesis is a Type I error.**
- B)** The power of a test is the probability of failing to reject the null hypothesis when it is false.
- C)** For a one-tailed test involving X , the null hypothesis would be $H_0: X = 0$, and the alternative hypothesis would be $H_A: X \neq 0$.

Explanation

Type I error is rejecting the null hypothesis when it is true. The power of a test is the probability of rejecting the null hypothesis when it is false. $H_A: X \neq 0$ indicates a two-tailed test, while $H_A: X < 0$ or $H_A: X > 0$ indicates a one-tailed test.

For Further Reference:

(Study Session 3, Module 11.1, LOS 11.b, 11.d)

CFA® Program Curriculum, Volume 1, page 612

CFA® Program Curriculum, Volume 1, page 615

Question #31 of 120

Question ID: 1207296

Questions 31 through 42 relate to Economics. (18 minutes)

A business cycle theory developed by applying utility theory and budget constraints to macroeconomic models is *most closely* associated with which school of economic thought?

- A)** Austrian.
- B) New Classical.**
- C)** New Keynesian.

Explanation

Real business cycle theory, which derives from applying utility theory and budget constraints to macroeconomic models, is associated with the New Classical school.

For Further Reference:

(Study Session 4, Module 15.1, LOS 15.c)

CFA® Program Curriculum, Volume 2, page 211

Question #32 of 120

Question ID: 1207299

A central bank is said to have credibility if:

- A) it issues inflation reports monthly.
- B) economic actors base decisions on the central bank's stated inflation targets.**
- C) it determines both the policy rate and the method for computing the inflation rate.

Explanation

If a central bank has credibility, economic actors come to believe the inflation rate will be near the central bank's target and factor this inflation rate into their decisions. Periodic inflation reports enhance the transparency of a central bank. A central bank that determines both the policy rate and the method for computing the inflation rate is said to have independence.

For Further Reference:

(Study Session 5, Module 16.2, LOS 16.j)

CFA® Program Curriculum, Volume 2, page 286

Question #33 of 120

Question ID: 1207303

Consider the following foreign exchange and interest rate information:

- Spot rate: 1.3382 USD/EUR.
- One year riskless USD rate = 2.5%.
- One year riskless EUR rate = 3.5%.

The one-year arbitrage-free forward exchange rate is *closest* to:

- A) 1.2391 USD/EUR.
- B) 1.3253 USD/EUR.**
- C) 1.3513 USD/EUR.

Explanation

Arbitrage-free forward rate = $1.3382 \text{ USD/EUR} \times (1.025 / 1.035) = 1.3253 \text{ USD/EUR}$.

For Further Reference:

(Study Session 5, Module 18.2, LOS 18.h)

CFA® Program Curriculum, Volume 2, page 423

Question #34 of 120

Question ID: 1207302

Other things equal, a country is *most likely* to have a current account deficit if it also has:

- A) a low savings rate.**
- B) a government budget surplus.
- C) a low rate of domestic investment.

Explanation

As shown by the fundamental macro relationship $(X - M) = (S - I) - (G - T)$, a current account deficit ($X < M$) is associated with a low savings rate, a high rate of domestic investment, or low government savings (i.e., a budget deficit).

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.i)

CFA® Program Curriculum, Volume 2, page 373

Question #35 of 120

Question ID: 1207293

Which of the following statements about monopolists is *most accurate*?

- A) Monopolists have imperfect information about the demand curve for their product.**
- B)** Without government intervention, monopolists will always earn economic profits.
- C)** A monopolist maximizes total revenue where marginal revenue equals marginal cost.

Explanation

Demand curves are not observable so a monopolist must search for the profit maximizing price. Because demand information is not perfect, a monopolist is a price searcher. The other statements are false. Although a monopolist can earn positive economic profits in the long run, they are not guaranteed; if average total costs exceed price, the monopolist will experience economic losses. A monopolist maximizes *profit*, not revenue, where marginal revenue equals marginal cost.

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.b, 13.d, 15.e)

CFA® Program Curriculum, Volume 2, page 64

CFA® Program Curriculum, Volume 2, page 68

CFA® Program Curriculum, Volume 2, page 68

Question #36 of 120

Question ID: 1207300

A central bank's ability to achieve its policy goals is *most likely* to be limited by available resources when which of the following actual rates is above its target rate?

- A)** Interest rate.
- B)** Inflation rate.
- C) Exchange rate.**

Explanation

With exchange rate targeting, a central bank's ability to increase the value of the domestic currency is limited by the amount of foreign reserves the country has available to buy its own currency in the foreign exchange market. While inflation targeting and interest rate targeting have limitations (e.g., liquidity trap conditions may exist, interest rates are bounded by zero), the central bank's resources are not typically a limitation.

For Further Reference:

(Study Session 5, Module 16.2, LOS 16.l, 16.n)

Question #37 of 120

Question ID: 1207297

Which of the following statements regarding the money supply and determination of short-term interest rates is *least accurate*?

- A) On balance, growth in real GDP tends to increase the transactional demand for money.
- B) If the short-term interest rate is greater than the equilibrium rate, there will be excess supply of real money balances.
- C) **An increase in the real money supply from an initial equilibrium situation will cause households and businesses to sell interest-bearing securities.**

Explanation

From an initial equilibrium, an increase in real money balances will leave households and businesses with more money than they wish to hold, so they will purchase interest-bearing securities, driving their prices up and yields down until a new equilibrium short-term rate is established.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.d)

CFA® Program Curriculum, Volume 2, page 269

Question #38 of 120

Question ID: 1207295

What are the *most likely* effects on aggregate demand in the current period of an increase in expected future incomes and of an increase in the money supply?

- A) **Both increase aggregate demand.**
- B) Both decrease aggregate demand.
- C) One increases aggregate demand and one decreases aggregate demand.

Explanation

An increase in expected future incomes will cause consumers to increase current expenditures (reduce current savings) in anticipation of the higher future incomes. An increase in the money supply will tend to decrease interest rates which will lead to increased consumer spending on durable goods and increased investment by businesses. Both effects increase aggregate demand.

For Further Reference:

(Study Session 4, Module 14.2, LOS 14.h)

CFA® Program Curriculum, Volume 2, page 148

Question #39 of 120

Question ID: 1207298

Incorrect production decisions are *most likely* to occur when the inflation rate is:

- A) lower than expected only.
- B) higher than expected only.
- C) either higher or lower than expected.**

Explanation

Either higher-than-expected or lower-than-expected inflation can cause producers to misinterpret unexpected changes in the price level as signals of increases or decreases in demand, and produce more or less than the equilibrium quantity of output.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.g)

CFA® Program Curriculum, Volume 2, page 279

Question #40 of 120

Question ID: 1207301

The source of comparative advantage, according to the Heckscher- Ohlin model of international trade, is each country's:

- A) labor productivity.
- B) available natural resources.
- C) relative amounts of labor and capital.**

Explanation

In the Heckscher-Ohlin model, the source of comparative advantage is the relative amounts of labor and capital that are available in each country. Countries with more capital available relative to labor available will have a comparative advantage in producing capital-intensive goods, while countries with more labor available relative to capital will have a comparative advantage in labor-intensive goods.

For Further Reference:

(Study Session 5, Module 17.1, LOS 17.d)

CFA® Program Curriculum, Volume 2, page 350

Question #41 of 120

Question ID: 1207294

Under which market structure is the profit maximizing strategy to produce the quantity of output for which the price is equal to marginal cost?

- A) Monopoly.
- B) Perfect competition.**
- C) Monopolistic competition.

Explanation

Firms' demand curves are perfectly elastic (horizontal) in a market characterized as perfect competition, so that marginal revenue is equal to price and a firm maximizes profit by producing the output quantity at which marginal cost equals price. In monopoly markets or under monopolistic competition, firm demand curves are downward sloping so that marginal revenue is less than price.

For Further Reference:

(Study Session 4, Modules 13.1, 13.2, 13.4, LOS 13.f)

Question #42 of 120

Question ID: 1207292

Which of the following statements about elasticity is *least accurate*?

- A) Both demand and supply are more elastic in the long run than in the short run.
- B) When demand is inelastic, an increase in price will cause a decrease in the total expenditure on a good.**
- C) When the price of a product increases, consumers will reduce their consumption by a larger amount in the long run than in the short run.

Explanation

If demand is inelastic, the percentage change in quantity demanded is smaller than the percentage change in price; quantity demanded is relatively unresponsive to price changes. A price increase increases total expenditures on a good.

For Further Reference:

(Study Session 4, Module 12.1, LOS 12.a)

CFA® Program Curriculum, Volume 2, page 9

Question #43 of 120

Question ID: 1207307

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

A U.S. GAAP reporting company invests \$50 million in a bond portfolio yielding 4% with an average maturity of seven years. After one year, interest rates have fallen by 50 basis points. The company will report the highest retained earnings if the securities in the portfolio are classified as:

- A) held-to-maturity.
- B) available-for-sale.
- C) trading securities.**

Explanation

The trading securities classification includes the unrealized gain from the bond in net income, which increases retained earnings. Unrealized gains on available-for-sale securities are reported as other comprehensive income for the period and are recorded in accumulated other comprehensive income, a component of owner's equity. Unrealized gains on held-to-maturity securities are not reported on the financial statements.

For Further Reference:

(Study Session 7, Module 22.6, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

Question #44 of 120

Question ID: 1207309

Kimberwick Technologies reported the following information for the year ending December 31.

Data	
Net sales	50,000
Cash expenses	3,250
Cash inputs	17,000
Cash taxes	7,000
Increase in receivables	500
Depreciation expense	1,000
Cash flow from investing	-5,000
Cash flow from financing	-4,250

If the cash balance increased \$13,000 over the year, cash flow from operations (CFO) is *closest* to:

- A) \$21,250.
- B) \$21,750.
- C) **\$22,250.**

Explanation

The easiest way to calculate CFO here is total cash flow – cash flow from investing – cash flow from financing = \$13,000 + 5,000 + 4,250 = \$22,250. Alternatively, CFO = \$50,000 – 3,250 – 17,000 – 7,000 – 500 = \$22,250.

For Further Reference:

(Study Session 7, Module 23.2, LOS 23.f)

CFA® Program Curriculum, Volume 3, page 199

Question #45 of 120

Question ID: 1207312

A company has a cash conversion cycle of 80 days. If the company's average receivables turnover increases from 11 to 12, the company's cash conversion cycle:

- A) **decreases by approximately 3 days.**
- B) increases by approximately 3 days.
- C) decreases by approximately 1 day.

Explanation

Cash conversion cycle (CCC) = days of sales outstanding + days of inventory on hand – number of days of payables. Days of sales outstanding = $365 / \text{receivables turnover}$ = $365 / 11 = 33.18$; $365 / 12 = 30.42$. This means the CCC decreases by 2.76 days.

For Further Reference:

(Study Session 7, Module 24.2, LOS 24.b)

CFA® Program Curriculum, Volume 3, page 256

Question #46 of 120

Question ID: 1207304

Which of the following statements about a United States public corporation's annual reports, SEC filings, and press releases is *most accurate*?

- A) Annual and quarterly SEC filings must be audited.
- B) Interim SEC filings typically update the major financial statements and footnotes.**
- C) Annual reports to shareholders are typically the most factual and objective source of information about a company.

Explanation

Besides the annual SEC filings, an analyst should examine a company's quarterly or semiannual filings. These interim filings typically update the major financial statements and footnotes, but are not necessarily audited. Annual reports to shareholders and press releases are written by management and are often viewed as public relations or sales materials.

For Further Reference:

(Study Session 6, Module 19.2, LOS 19.e)

CFA® Program Curriculum, Volume 3, page 31

Question #47 of 120

Question ID: 1207320

Which of the following is *most likely* a motivation for a company's management to issue low-quality financial reports?

- A) Management has provided optimistic earnings guidance.**
- B) Oversight provided by the board of directors is weak or inadequate.
- C) Accounting principles permit a wide range of acceptable treatments and estimates.

Explanation

Meeting or exceeding its own earnings guidance is a possible motivation for management to issue low-quality financial reports. Inadequate board oversight and wide ranges of acceptable accounting treatments are more appropriately viewed as opportunities for issuing low-quality financial reports.

For Further Reference:

(Study Session 9, Module 29.1, LOS 29.d, LOS 29.e)

CFA® Program Curriculum, Volume 3, page 557

CFA® Program Curriculum, Volume 3, page 558

Question #48 of 120

Question ID: 1207314

Haltata Turf & Sod currently uses the first in, first out (FIFO) method to account for inventory. Due to significant tax-loss carryforwards, the company has an effective tax rate of zero. Prices are rising and inventory quantities are stable. If the company were to use last in, first out (LIFO) instead of FIFO:

- A) net income would be lower and cash flow would be higher.
- B) cash flow would remain the same and working capital would be lower.**

C) gross margin would be higher and stockholder's equity would be lower.

Explanation

In the absence of taxes, there is no difference in cash flow between LIFO and FIFO. In addition, using LIFO would result in lower working capital (inventory is lower). Using LIFO would result in lower net income because of a *lower* gross margin (cost of goods sold is higher).

For Further Reference:

(Study Session 8, Module 25.5, LOS 25.k, 25.l)

CFA® Program Curriculum, Volume 3, page 340

CFA® Program Curriculum, Volume 3, page 377

Question #49 of 120

Question ID: 1207321

Which of the following effects is *most likely* to occur when using ratio screens for high dividend yield stocks and low P/E stocks, respectively?

High dividend yield

Low P/E ratios

- | | |
|--|--------------------------------------|
| A)
Include too many financial services firms | Exclude too many growth firms |
| B) Exclude too many financial services firms | Include too many growth firms |
| C) Include too many financial services firms | Include too many growth firms |

Explanation

Screening for high dividend yield stocks will likely include a disproportionately high number of financial services firms as such firms typically have higher dividend payouts. A screen to identify firms with low P/E ratios will likely exclude growth firms from the sample as high expected earnings growth leads to high P/Es.

For Further Reference:

(Study Session 9, Module 30.2, LOS 30.d)

CFA® Program Curriculum, Volume 3, page 624

Question #50 of 120

Question ID: 1207305

The five steps required for a company to record revenue on a long-term contract are *least likely* to include:

- A)** identifying a customer contract.
- B) receiving proportional payments.**
- C)** identifying separate performance obligations in the contract.

Explanation

Receipt of payments is not one of the required steps described by accounting standards to recognize revenue for a long-term contract.

For Further Reference:

(Study Session 7, Module 21.2, LOS 21.b)

CFA® Program Curriculum, Volume 3, page 79

CFA® Program Curriculum, Volume 3, page 99

Question #51 of 120

Question ID: 1207308

A company's investments in marketable securities include a 3-year tax-exempt bond measured at amortized cost and a 5-year government bond measured at fair value through other comprehensive income. On its income statement, the company should report the coupon interest received from:

- A) both of these securities.**
- B) neither of these securities.**
- C) only one of these securities.**

Explanation

Interest and dividends received are reported as income, regardless of the balance sheet classification of marketable securities.

For Further Reference:

(Study Session 7, Module 22.6, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

Question #52 of 120

Question ID: 1207315

Acme Corp. purchased a new stamping machine for \$100,000, paid \$10,000 for shipping, and paid \$5,000 to have it installed in their plant. Based on an estimated salvage value of \$25,000 and an economic life of six years, the difference between straight-line depreciation and double-declining balance depreciation in the second year of the asset's life is *closest* to:

- A) \$7,220.**
- B) \$10,556.**
- C) \$16,666.**

Explanation

Straight line depreciation is $(100,000 + 10,000 + 5,000 - 25,000) / 6 = 15,000$ each year. Double-declining balance depreciation in the second year is: $115,000 (2/3)(1/3) = 25,556$. The difference is \$10,556. Remember that salvage value is not part of the declining balance calculation. and

For Further Reference:

(Study Session 7, Module 21.3, LOS 21.d)

CFA® Program Curriculum, Volume 3, page 398

Question #53 of 120

Question ID: 1207310

How will a firm's operating cash flow be affected by a decrease in accounts receivable and by an increase in accounts payable?

- A) Both will increase operating cash flow.**
- B) Both will decrease operating cash flow.**
- C) One will increase operating cash flow and one will decrease operating cash flow.**

Explanation

A decrease in the accounts receivable amount on the balance sheet indicates that cash collections exceed revenues (sales). This increases operating cash flow because receivables are being collected. An increase in the accounts payable amount on the balance sheet indicates that purchases from suppliers exceed cash payments. This increases operating cash flow because the cash was not used to pay the suppliers.

For Further Reference:

(Study Session 7, Module 23.2, LOS 23.f)

CFA® Program Curriculum, Volume 3, page 199

Question #54 of 120

Question ID: 1207313

From the point of view of a financial analyst, when evaluating companies that use different inventory cost assumptions, in a period of:

- A) stable prices, LIFO inventory is preferred to FIFO inventory.**
- B) decreasing prices, FIFO inventory is preferred to LIFO inventory.**
- C) increasing prices, FIFO cost of sales is preferred to LIFO cost of sales.**

Explanation

The most useful estimates of inventory and cost of sales are those that best approximate current cost. Whether prices are increasing or decreasing, FIFO provides a better estimate of inventory values, and LIFO provides a better estimate of cost of sales. If prices are stable, there is no difference between LIFO and FIFO estimates of inventory or cost of sales.

For Further Reference:

(Study Session 8, Module 25.2, LOS 25.d)

CFA® Program Curriculum, Volume 3, page 320

Question #55 of 120

Question ID: 1207319

An IFRS reporting company and a U.S. GAAP reporting company have entered into identical leases of significant value, as the lessees. Compared to the impact of reporting the lease under IFRS, reporting the lease under U.S. GAAP as an operating lease will:

- A) add less assets to their balance sheet.**
- B) decrease operating cash flows by a greater amount.**
- C) decrease financing cash flows by a greater amount.**

Explanation

An IFRS company will report the interest portion of the lease payment and depreciation of the lease asset separately as expenses on the income statement. Only the interest payment represents an operating cash outflow (and may be treated as a financing cash outflow). For an operating lease, a U.S. GAAP company will report the entire lease payment as an expense on their income statement, and as an operating cash outflow. Both companies will report the same lease asset on their balance sheets. The lease will not affect the U.S. GAAP company's cash flow from financing. The principal portion of the lease payment will be reported as a financing cash outflow by the IFRS company.

For Further Reference:

(Study Session 8, Module 28.4, LOS 28.g)

CFA® Program Curriculum, Volume 3, page 510

Question #56 of 120

Question ID: 1207316

Longboat, Inc., sold a luxury passenger boat from its inventory on December 31 for \$2,000,000. It is estimated that Longboat will incur \$100,000 in warranty expenses during its 5-year warranty period. Longboat's tax rate is 30%. To account for the tax implications of the warranty obligation prior to incurring warranty expenses, Longboat should:

- A) record a deferred tax asset of \$30,000.**
- B) record a deferred tax liability of \$30,000.**
- C) make no entry until actual warranty expenses are incurred.**

Explanation

Warranty expense should be recorded when the inventory item covered by the warranty is sold. A deferred tax asset is created when warranty expenses are accrued on the financial statements but are not deductible on the tax returns until the warranty claims are paid. The full amount of the obligation, \$100,000, is recorded as an expense, with a deferred tax asset of \$30,000. Note that a deferred tax asset results when taxable income is more than pretax income and the difference is likely to reverse (warranty will be paid) in future years.

For Further Reference:

(Study Session 8, Module 27.3, LOS 27.d)

CFA® Program Curriculum, Volume 3, page 454

Question #57 of 120

Question ID: 1207306

During 20X3, Rory, Inc., reported net income of \$15,000 and had 2,000 shares of common stock outstanding for the entire year. Rory also had 2,000 shares of 10%, \$50 par value preferred stock outstanding during 20X3. During 20X1, Rory issued 100, \$1,000 par, 6% bonds for \$100,000. Each of the bonds is convertible to 50 shares of common stock. Rory's tax rate is 40%. Assuming these bonds are dilutive, 20X3 diluted EPS for Rory is *closest* to:

- A) \$0.71.**
- B) \$1.23.**
- C) \$2.50.**

Explanation

$$\text{Diluted EPS} = [\text{NI} - \text{preferred dividends} + \text{convertible interest } (1 - t)] / [\text{weighted average shares} + \text{convertible debt shares}]$$

$100(1,000)(6\%)(1 - 0.4) = \$3,600$; convertible debt shares = $50(100) = 5,000$

$$\frac{\$15,000 - \$10,000 + \$3,600}{2,000 + 5,000} = \$1.23$$

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g)

CFA® Program Curriculum, Volume 3, page 99

Question #58 of 120

Question ID: 1207318

A firm issues a 4-year semiannual-pay bond with a face value of \$10 million and a coupon rate of 10%. The market interest rate is 11% when the bond is issued. The balance sheet liability at the end of the first semiannual period is *closest* to:

- A) \$9,650,700.
- B) \$9,683,250.
- C) **\$9,715,850.**

Explanation

The initial liability is the amount received from the creditor, not the par value of the bond.

$$N = 8; I/Y = 11/2 = 5.5; PMT = 500,000; FV = 10,000,000; CPT \rightarrow PV = \$9,683,272.$$

The interest expense is the effective interest rate (the market rate at the time of issue) times the balance sheet liability. $\$9,683,272 \times 0.055 = \$532,580$.

The value of the liability will change over time and is a function of the initial liability, the interest expense and the actual cash payments. In this case, it increases by the difference between the interest expense and the actual cash payment: $\$532,580 - \$500,000 = \$32,580 + \$9,683,272 = \$9,715,852$. *Tip:* Knowing that the liability will increase is enough to select choice C without performing this last calculation. Entering $N = 7$ and solving for PV also produces \$9,715,852.

For Further Reference:

(Study Session 8, Module 28.2, LOS 28.b)

CFA® Program Curriculum, Volume 3, page 494

Question #59 of 120

Question ID: 1207311

The presentation format of balance sheet data that standardizes the first-year values to 1.0 and presents subsequent years' amounts relative to 1.0 is:

- A) an indexed balance sheet.
- B) a vertical common-size balance sheet.
- C) **a horizontal common-size balance sheet.**

Explanation

On a horizontal common-size balance sheet, the divisor is the first-year values so they are all standardized to 1.0 by construction. Trends in the values of these items as well as the relative growth in these items are readily apparent. A vertical common-size balance

sheet expresses all balance sheet accounts as a percentage of total assets and does not standardize the initial year.

For Further Reference:

(Study Session 7, Module 24.1, LOS 24.a)

CFA® Program Curriculum, Volume 3, page 237

Question #60 of 120

Question ID: 1207317

For 20X1, Belcher Motors reported a decrease in its deferred tax liabilities, a decrease in its deferred tax assets, and an increase in its valuation allowance. To an analyst, this would *most likely* suggest that the company has:

- A) decreased its estimate of future profitability.**
- B) increased the estimated useful life of some capitalized assets.
- C) increased its estimate of the period over which unearned revenue will be recognized.

Explanation

The increase in the valuation allowance tells us that the company has decreased its estimate of its future profitability and thus its ability to realize the benefits of its deferred tax assets. A longer period for recognition of unearned revenue would not affect the temporary differences reflected in deferred tax assets. Increasing the estimate of assets' useful lives would tend to slow financial statement depreciation relative to depreciation for tax, which would increase deferred tax liability going forward, other things constant. Decreases in the carrying values of both a DTL and a DTA may reflect a decrease in the tax rate.

For Further Reference:

(Study Session 8, Module 27.5, LOS 27.g)

CFA® Program Curriculum, Volume 3, page 469

Question #61 of 120

Question ID: 1207332

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

At the beginning of the year, Breidel Company changes its inventory accounting method (for both financial and tax reporting) from first in first out to average cost. Assuming an environment of increasing prices, how will this accounting change affect Breidel's forecasts of its net cash position?

- A) No effect because this accounting change does not affect cash flows.**
- B) Less net cash in both the short-term forecast and the long-term forecast.
- C) No effect on the short-term forecast but greater net cash in the long-term forecast.**

Explanation

Changing the inventory accounting method has no immediate cash flow effects and therefore should not change a firm's short-term forecast (typically 4 to 6 weeks) of its net cash position. However, because the average cost inventory method will result in lower gross profit compared to FIFO, it will also result in decreased taxes. The firm's long-term forecast (typically 3 to 5 years) of its net cash position should reflect a decrease in cash outflows for taxes, and consequently greater net cash in future periods.

For Further Reference:

(Study Session 8, Module 25.5, LOS 25.I, Study Session 11, Module 35.1, LOS 35.d)

Question #62 of 120

Question ID: 1207325

Which of the following statements is *most accurate* regarding the net present value (NPV) and internal rate of return (IRR) capital budgeting methods?

- A) NPV assumes that cash flows can be reinvested at the project's IRR.
- B) IRR assumes the cash flows are reinvested at the project's cost of capital.
- C) NPV assumes the cash flows can be reinvested at the project's cost of capital.**

Explanation

The NPV method implicitly assumes that cash flows can be reinvested at the project's cost of capital, while the IRR method assumes reinvestment at the computed IRR.

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.e)

CFA® Program Curriculum, Volume 4, page 58

Question #63 of 120

Question ID: 1207328

When using the CAPM to estimate the cost of common equity for a company in a developing country, an analyst should *most* appropriately:

- A) add a country risk premium to the market risk premium.**
- B) add the sovereign yield spread to the CAPM cost of common equity.
- C) make no adjustments because country risk is reflected in the equity's beta.

Explanation

The appropriate method for estimating the cost of equity for a firm in a developing market is to add a country risk premium (CRP) to the market risk premium, so the revised CAPM equation becomes: $k_{ce} = R_F + \beta[E(R_{MKT}) - R_F + CRP]$. The CRP is the sovereign yield spread (between yields on the country's government bonds and a developed country's government bonds) adjusted for the volatility of the developing country's equity market. An alternative approach is to add the CRP (not the sovereign yield spread) to the cost of equity as calculated from the CAPM.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.j)

CFA® Program Curriculum, Volume 4, page 101

Question #64 of 120

Question ID: 1207333

The type of short-term financing for which the financing cost is *most closely* tied to the creditworthiness of a firm's customers is:

- A) factoring.**
- B) issuing commercial paper.**
- C) an uncommitted line of credit.**

Explanation

Factoring refers to the sale of receivables without recourse; that is, the risk that the firm's customers will not pay, or will not pay in a timely manner, is borne by the factor, who purchases the receivables. Thus, the amount the factor will pay per dollar of receivables is lower (higher discount or interest rate) if the credit quality of the firm's credit customers is lower.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.g)

CFA® Program Curriculum, Volume 4, page 189

Question #65 of 120

Question ID: 1207330

Smith Company's earnings per share are more sensitive to changes in operating income than are those of Jones Company. This implies that Smith Company has a higher degree of:

- A) total leverage.**
- B) financial leverage.**
- C) operating leverage.**

Explanation

The degree of financial leverage (DFL) is the percent change in earnings per share for a given percent change in operating income. The degree of operating leverage (DOL) is the percent change in operating income for a given percent change in sales. The degree of total leverage (DTL) is the percent change in earnings per share for a given percent change in sales, and is the product of DOL and DFL. Based on the information given, Smith has a higher DFL than Jones, but we cannot conclude that Smith has a higher DTL than Jones.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.b)

CFA® Program Curriculum, Volume 4, page 128

Question #66 of 120

Question ID: 1207329

If flotation costs are treated correctly in calculating the net present value of a project that will begin in the current period, the flotation costs are *most likely*:

- A) included in the initial outlay.**
- B) reflected in the discount rate used for the project.**
- C) included in the cost of the capital raised.**

Explanation

The correct treatment of flotation costs according to the CFA Curriculum is to include their cost in the initial cash outflow of the project. Since flotation costs are included in the initial outlay, they decrease the NPV by an amount that is unaffected by the discount rate, and the discount rate and cost of capital are not adjusted for flotation costs.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.1)

CFA® Program Curriculum, Volume 4, page 105

Question #67 of 120

Question ID: 1207326

Thompson Products has seen its marginal tax rate increase from 28% to 34% over the last two years and believes the change is permanent. The effects of this change on Thompson's current WACC and on its financial leverage over time are *most likely*:

- A) an increase in both.
- B) a decrease in both.
- C) an increase in one and a decrease in the other.

Explanation

An increase in the tax rate will reduce Thompson's after-tax cost of debt (other things equal) and therefore reduce its WACC. With a relatively lower cost of debt the firm will likely change its capital structure to include more debt and less equity (i.e., to increase its financial leverage).

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.b)

CFA® Program Curriculum, Volume 4, page 79

Question #68 of 120

Question ID: 1207323

Which of the following investments is *least likely* to be evaluated using traditional capital budgeting analysis? Investing:

- A) \$100 million for the research and development of new paint products.
- B) \$200 million in equipment to reduce carbon output rather than pay daily fines.
- C) \$75 million to expand production and sales into a neighboring country.

Explanation

Some research and development projects may involve such a high degree of risk that they are very difficult to evaluate using traditional capital budgeting methods. Replacement projects and mandatory environmental projects can typically be evaluated with the usual capital budgeting techniques.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.a)

CFA® Program Curriculum, Volume 4, page 48

Question #69 of 120

Question ID: 1207331

Daley Company sells its output for \$15 per unit. Daley's variable costs, including taxes, are \$10 per unit and its breakeven quantity of sales is 30,000 units. Daley's annual fixed costs are \$50,000 for interest and \$100,000 for rent. If Daley sells 35,000 units in a year, its

net income will be:

- A) \$15,000.
- B) \$25,000.**
- C) \$35,000.

Explanation

The contribution of each unit sold to covering fixed costs is $\$15 - \$10 = \$5$. Because selling 30,000 units just covers fixed costs, each additional unit sold produces a profit of \$5. Profit is $(\$35,000 - \$30,000) \times \$5 = \$25,000$.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.d)

CFA® Program Curriculum, Volume 4, page 143

Question #70 of 120

Question ID: 1207322

Reviewing the performance and independence of board members is a responsibility of:

- A) the audit committee.
- B) the nominations committee.**
- C) the compensation committee.

Explanation

The nominations committee is responsible for reviewing the performance, independence, skills, and experience of board members, as well as for recruiting new board members and creating board nomination policies.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.f)

CFA® Program Curriculum, Volume 4, page 21

Question #71 of 120

Question ID: 1207327

A company's pretax cost of fixed-rate debt capital equals the company's new debt:

- A) coupon rate.
- B) current yield.
- C) yield to maturity.**

Explanation

The yield to maturity correctly measures the pretax cost of debt financing.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.f)

CFA® Program Curriculum, Volume 4, page 84

Question #72 of 120

Question ID: 1207324

A company is evaluating the following capital projects for investment over the next two years:

- Two new machines with costs of \$4 million each.
- Computer software upgrade with a cost of \$1 million.
- Multi-year replacement of two aging machines involving an investment of \$4.5 million for the first machine and another \$4.5 million for the second machine if projected savings from the first machine are realized.

All of these projects have positive net present values and the available budget is \$10 million. The company should accept:

- A) all of these projects.**
- B) those projects with the highest expected rates of return over the 2-year capital budgeting period.**
- C) those projects with the highest present value of expected future cash flows relative to required investment.**

Explanation

When a capital budget is limited, the available funds should be allocated among projects that are expected to increase the value of the firm by the greatest amount. The increase in firm value can be measured by the project's NPV. The firm should calculate the NPV of each project and find the mix that maximizes the increase in firm value and does not exceed the funds available for investment.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.c)

CFA® Program Curriculum, Volume 4, page 51

Question #73 of 120

Question ID: 1207342

Questions 73 through 85 relate to Equity Investments. (19.5 minutes)

An analyst determines that a company has a return on equity of 16% and pays 40% of its earnings in dividends. If the firm recently paid a \$1.50 dividend and the stock is selling for \$40, what is the required rate of return on the stock if it is priced according to the dividend discount model?

- A) 9.6%.**
- B) 10.2%.**
- C) 13.7%.**

Explanation

$$g = (1 - \text{payout})(\text{ROE}) = (1 - 0.40)(16\%) = 9.6\%$$

$$k = \frac{\$1.50(1.096)}{\$40} + 0.096 = 13.7\%$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #74 of 120

Question ID: 1207334

An investor buys a stock for \$50. The initial margin requirement is 50%, and the maintenance margin requirement is 25%. The price below which the investor would receive a margin call would be:

- A) \$25.00.
- B) \$33.33.**
- C) \$37.50.

Explanation

The following formula indicates the stock price that will trigger a margin call:

$$\begin{aligned} \text{long} &= \frac{(\text{original price})(1 - \text{initial margin \%})}{1 - \text{maintenance margin}} \\ &= \frac{(\$50)(1 - 0.50)}{1 - 0.25} \\ &= \$33.33 \end{aligned}$$

For Further Reference:

(Study Session 12, Module 36.2, LOS 36.f)

CFA® Program Curriculum, Volume 4, page 243

Professor's Note

An intuitive way to solve minimum margin problems for equity accounts is based on the fact that while the margin amount changes with stock price changes after purchase, the loan amount does not. Stock price is \$50, loan is \$25, when the margin is 25% the loan must be 75% (of share price). $\$25 / 0.75 = \33.33 .

Question #75 of 120

Question ID: 1207335

A securities market exhibits operational efficiency if it offers:

- A) low transaction costs.**
- B) prices that respond rapidly to new information.
- C) rates of return that are proportional to risk on average.

Explanation

Operational efficiency refers to low transactions costs in a securities market. Informational efficiency means prices change rapidly to reflect new information without predictable bias and rates of return are, on average, proportional to risk.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.k)

CFA® Program Curriculum, Volume 4, page 260

Question #76 of 120

Question ID: 1207337

A stock index consists of two stocks. As of January 1:

- Company A has 50 shares outstanding valued at \$2 each.
- Company B has 10 shares outstanding valued at \$10 each.
- The price-weighted index is 6, and the value-weighted index is 100.

On June 30, the price of Company A's stock has increased to \$4 per share. Effective the morning of July 1, Company B's stock splits two-for-one and is priced at \$5. The opening values of the price-weighted index and the value-weighted index on July 1 are:

	<u>Price-weighted</u>	<u>Value-weighted</u>
A) 7		150
B) 7		125
C) 4.5		150

Explanation

Price-weighted index = $\frac{4+10}{2} = 7$. A price-weighted index is not affected by a split.

The divisor is adjusted to account for the price change.

$$\text{Value-weighted index} = 100 \times \frac{\$4(50) + \$5(20)}{\$2(50) + \$10(10)} = 150$$

For Further Reference:

(Study Session 12, Module 37.1, LOS 37.e)

CFA® Program Curriculum, Volume 4, page 285

Question #77 of 120

Question ID: 1207344

Pam Robers, CFA, is performing a valuation analysis on the common stock of Allstare Inc. The stock's beta is 1.1, the risk-free rate is 5%, and the market risk premium is expected to be 8%. Allstare's ROE is expected to be constant at 18%, and its dividend payout ratio has been fairly constant over time at 40%. The forward-earnings multiplier that Robers should use to estimate the current value of the shares is *closest* to:

- A) 7.
- B) 13.
- C) 20.

Explanation

$$\text{Required return} = 5\% + 1.1(8\%) = 13.8\%$$

$$\text{Sustainable growth} = 18\%(1 - 0.4) = 10.8\%$$

$$P_0/E_1 = \frac{0.40}{0.138 - 0.108} = 13.33$$

For Further Reference:

(Study Session 13, Modules 41.2, 41.3, LOS 41.e, 41.j)

Question #78 of 120

Question ID: 1207343

Pat McCoy, CFA, is analyzing a technology firm that has experienced annual earnings growth of 12%. McCoy does not expect the firm to begin paying dividends on its common shares in the foreseeable future. To estimate the value of this firm's common shares, McCoy should *most appropriately* use:

- A) a two-stage DDM.
- B) a free cash flow model.**
- C) a Gordon growth model.

Explanation

Free cash flow-based valuation techniques are appropriate for valuing shares of a firm that does not pay dividends. The Gordon growth model and two-stage dividend discount model are appropriate for valuing shares of dividend-paying firms.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.h)

CFA® Program Curriculum, Volume 4, page 463

Question #79 of 120

Question ID: 1207341

Over the past few years, the companies in an industry have experienced positive but decreasing profitability and growth rates. The companies have begun to compete intensely with each other and customers switch frequently among brands. This industry's life-cycle stage is *most accurately* described as:

- A) growth.
- B) maturity.
- C) shakeout.**

Explanation

The shakeout industry life-cycle stage is characterized by slowing (but still positive) growth, intense competition, and declining profitability, as demand begins to approach market saturation. In contrast, the decline industry stage is characterized by negative growth. The lack of brand loyalty among customers suggests the industry has not yet reached the mature stage.

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.h)

CFA® Program Curriculum, Volume 4, page 414

Question #80 of 120

Question ID: 1207345

An analyst gathered the following data about a company:

- The historical earnings retention rate of 40% is projected to continue into the future.
- The sustainable ROE is 12%.
- The stock's beta is 1.2.
- The nominal risk-free rate is 6%.
- The expected market return is 11%.

If the analyst believes next year's earnings will be \$4 per share, what value should be placed on this stock?

- A) \$22.24.
- B) \$33.32.**
- C) \$45.45.

Explanation

Dividend payout = 1 – earnings retention rate = 1 – 0.4 = 0.6

$R_S = R_f + \beta(R_M - R_f) = 0.06 + 1.2(0.11 - 0.06) = 0.12$

$g = (\text{retention rate})(\text{ROE}) = 0.4(0.12) = 0.048$

$$P/E = \frac{\text{dividend payout ratio}}{k-g} = \frac{0.6}{0.12-0.048} = 8.33$$

Price = E(P/E) = \$4(8.33) = \$33.32

For Further Reference:

(Study Session 13, Modules 41.2, 41.3, LOS 41.e, 41.j)

CFA® Program Curriculum, Volume 4, page 453

CFA® Program Curriculum, Volume 4, page 468

Question #81 of 120

Question ID: 1207340

Which of the following classifications of firms is *least likely* to comprise cyclical firms?

- A) Housing.
- B) Technology.
- C) Telecommunications.**

Explanation

Both technology and housing firms tend to be quite cyclical, that is, their profits are very sensitive to changes in overall growth. The profits of telecommunications firms, on the other hand, are less economically sensitive.

For Further Reference:

(Study Session 13, Module 40.1, LOS 40.c)

CFA® Program Curriculum, Volume 4, page 390

Question #82 of 120

Question ID: 1207338

The assertion that investors, analysts, and portfolio managers exhibit psychological tendencies that cause them to make systematic errors is *most* consistent with:

- A) behavioral finance.**
- B) weak-form market efficiency.
- C) fundamental analysis.

Explanation

Behavioral finance studies how market anomalies can arise from psychological traits that affect investor behavior and cause investors to make systematic errors such as exiting profitable positions too soon and holding unprofitable positions too long.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.g)

CFA® Program Curriculum, Volume 4, page 338

Question #83 of 120

Question ID: 1207336

An investor purchases 1,000 shares of each of the stocks in a price- weighted index at their closing prices (ignore transactions costs). On a total return basis, if the index stocks remain the same, this portfolio will *most likely*:

- A) perform exactly like the index over time.**
- B) outperform the index over time.**
- C) underperform the index over time.

Explanation

Total return includes dividend yield. Because dividends are not included in the performance of the index itself, the portfolio will outperform the index by the amount of the dividend yield.

For Further Reference:

(Study Session 12, Module 37.1, LOS 37.b)

CFA® Program Curriculum, Volume 4, page 281

Question #84 of 120

Question ID: 1207346

Among valuation models, the difficulty of estimating a required rate of return is *most likely* to be a disadvantage of using:

- A) a Gordon growth model.**
- B) an asset-based valuation model.
- C) an enterprise value multiplier model.

Explanation

One of the disadvantages of present value models such as the Gordon growth model is that the required rate of return on equity must be estimated. Neither an enterprise value multiplier model nor an asset valuation model requires an explicit estimate of the required rate of return.

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.m)

CFA® Program Curriculum, Volume 4, page 448

Question #85 of 120

Question ID: 1207339

An investor owns preference shares which stipulate that any dividend for the current period, as well as all past dividends, must be fully paid before a common stock dividend may be paid. This security is *best* described as:

- A) full-pay preferred.
- B) restricted preferred.
- C) **cumulative preferred.**

Explanation

Cumulative preferred shares stipulate that all preferred dividends must be paid before common stock dividends can be paid.

For Further Reference:

(Study Session 13, Module 39.1, LOS 39.a)

CFA® Program Curriculum, Volume 4, page 357

Question #86 of 120

Question ID: 1207348

Questions 86 through 99 relate to Fixed Income. (21 minutes)

A 7.5% coupon, semiannual-pay, five-year bond has a yield to maturity of 6.80%. Over the next year, if the bond's yield to maturity remains unchanged, its price will:

- A) increase.
- B) **decrease.**
- C) remain unchanged.

Explanation

Because the coupon rate is greater than its yield to maturity, the bond price is at a premium to par value. If the yield remains unchanged, the price will decrease toward par value along its constant-yield price trajectory.

For Further Reference:

(Study Session 14, Module 44.3, LOS 44.b)

CFA® Program Curriculum, Volume 5, page 113

Question #87 of 120

Question ID: 1207359

The type of credit risk that is most directly reflected in a bond's rating is:

- A) default risk.
- B) downgrade risk.
- C) credit spread risk.

Explanation

Default risk is the possibility that the issuer will fail to meet its obligations under the indenture, for which investors demand a premium above the return on a default-risk-free security. Bond ratings indicate default risk. Downgrade risk is the risk that a bond will be reclassified as a riskier security by a credit rating agency. Credit spread risk is the risk that the default risk premium on a bond can increase.

For Further Reference:

(Study Session 15, Module 47.1, LOS 47.a, 47.e)

CFA® Program Curriculum, Volume 5, page 302

CFA® Program Curriculum, Volume 5, page 315

Question #88 of 120

Question ID: 1207365

What effects will an increase in yield volatility have on the values of a putable bond and a callable bond?

- A) Both bonds will increase in value.
- B) Both bonds will decrease in value.
- C) **One bond will increase in value and the other will decrease.**

Explanation

A callable bond is made up of a straight bond and a written call option. An increase in volatility increases the value of the call option and decreases the value of the callable bond. On the other hand, a putable bond is made up of an option-free (or straight) bond and a long put option. An increase in volatility increases the value of the put option and therefore increases the value of the putable bond.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.f, Study Session 16, Module 49.3, LOS 49.k)

CFA® Program Curriculum, Volume 5, page 37

CFA® Program Curriculum, Volume 5, page 482

Question #89 of 120

Question ID: 1207347

Which of the following statements about debt securities is *least accurate*?

- A) Commercial paper is a short-term vehicle for corporate borrowing.
- B) A securitized bond is a security whose cash flows are linked to a pool of underlying loans or financial instruments.
- C) **A medium-term note (MTN) is a corporate bond with an original maturity of 2 to 10 years.**

Explanation

The name "medium-term note" does not imply anything about the original maturity of the security.

For Further Reference:

(Study Session 14, Module 43.2, LOS 43.g)

CFA® Program Curriculum, Volume 5, page 79

Question #90 of 120

Question ID: 1207356

Portfolio duration *most accurately* approximates the sensitivity of the value of a bond portfolio to:

- A) parallel shifts in the yield curve.**
- B)** increases in the slope of the yield curve.
- C)** decreases in the slope of the yield curve.

Explanation

Portfolio duration is an approximation of the price sensitivity of a portfolio to parallel shifts of the yield curve (yields for all maturities increase or decrease by equal amounts). Key rate duration may be used to estimate interest rate risk for non-parallel shifts in the yield curve.

For Further Reference:

(Study Session 15, Module 46.2, LOS 46.d, 46.f)

CFA® Program Curriculum, Volume 5, page 259

CFA® Program Curriculum, Volume 5, page 265

Question #91 of 120

Question ID: 1207360

A firm is said to have a top-heavy capital structure if a high percentage of its total capital is:

- A)** debt.
- B)** short-term debt.
- C) secured bank debt.**

Explanation

"Top-heavy" refers to a capital structure that includes a high percentage of secured bank debt. A firm with a top-heavy capital structure may be limited in its access to additional bank borrowing, which increases the likelihood of default if the firm encounters financial distress.

For Further Reference:

(Study Session 15, Module 47.2, LOS 47.j)

CFA® Program Curriculum, Volume 5, page 347

Question #92 of 120

Question ID: 1207355

An 8%, semiannual pay, option-free corporate bond that is selling at par has ten years to maturity. What is the approximate modified duration of the bond based on a 75 basis point change (up or down) in rates?

- A) 5.6.
- B) 6.8.
- C) 7.2.

Explanation

First calculate V_- and V_+ , the bond's value at 7.25% and 8.75% yields to maturity. The bond values are \$1,052.70 and \$950.69, respectively:

$N = 20$; $I/Y = 7.25 / 2 = 3.625$; $PMT = 40$; $FV = 1,000$; $CPT PV = -1,052.70$

$N = 20$; $I/Y = 8.75 / 2 = 4.375$; $PMT = 40$; $FV = 1,000$; $CPT PV = -950.69$

$$D = \frac{V_- - V_+}{2V_0(\Delta y)} = \frac{1,052.70 - 950.69}{2(1,000)(0.0075)} = 6.8.$$

For Further Reference:

(Study Session 15, Module 46.1, LOS 46.b)

CFA® Program Curriculum, Volume 5, page 248

Question #93 of 120

Question ID: 1207350

The current 4-year spot rate is 4% and the current 5-year spot rate is 5.5%. What is the 1-year forward rate in four years?

- A) 9.58%.
- B) 10.14%.
- C) 11.72%.

Explanation

$$4y1y = \frac{(1.055)^5}{(1.04)^4} - 1 = 0.1172$$

Note: $5(5.5) - 4(4) = 11.5\%$.

For Further Reference:

(Study Session 14, Module 44.4, LOS 44.j)

CFA® Program Curriculum, Volume 5, page 143

Question #94 of 120

Question ID: 1207351

A bond with nine years to maturity is quoted at an interpolated spread of +150 basis points. The benchmark yield for this bond is:

- A) a swap rate.
- B) a matrix rate.
- C) a government bond yield.

Explanation

Interpolated spreads (I-spreads) are spreads to swap rates.

For Further Reference:

(Study Session 14, Module 44.5, LOS 44.k)

CFA® Program Curriculum, Volume 5, page 147

Question #95 of 120

Question ID: 1207352

Compared to a recourse mortgage loan, a nonrecourse mortgage loan:

- A) is likely to have a higher interest rate.**
- B)** is less likely to be purchased by a government agency.
- C)** is issued with less strict standards for a borrower's credit history.

Explanation

In the event of a default, a borrower with a nonrecourse loan is not liable for the loan amount, the property is simply returned to the lender. With a nonrecourse loan, the lender has more risk in the event of default so interest rates will be higher. Even without recourse, the creditworthiness of a borrower is a primary concern to originators of residential mortgage loans.

For Further Reference:

(Study Session 14, Module 45.1, LOS 45.d)

CFA® Program Curriculum, Volume 5, page 190

Question #96 of 120

Question ID: 1207353

A 10-year note issued by Gaullic Finance will be paid from a bankruptcy-remote pool of Gaullic's balance sheet assets. These notes are *best* described as:

- A) covered bonds.**
- B)** securitized bonds.
- C)** asset-backed bonds.

Explanation

Covered bonds are an obligation of the corporation that issues them, but their interest and principal payments are provided by a pool of assets that are legally recognized as bankruptcy remote. They are different from securitized bonds (i.e., asset-backed securities), which are issued by a special purpose entity to which the underlying assets are sold.

For Further Reference:

(Study Session 14, Modules 42.1, 45.1, LOS 42.d, 45.b)

CFA® Program Curriculum, Volume 5, page 21

CFA® Program Curriculum, Volume 5, page 181

Question #97 of 120

Question ID: 1207354

Extension in an agency residential mortgage-backed security is *most likely* to result from:

- A) a decrease in interest rates.
- B) exhaustion of a support tranche.
- C) **slower-than-expected prepayments.**

Explanation

An agency RMBS is said to extend when prepayments of the underlying mortgages are slower than expected. A decrease in interest rates would tend to accelerate prepayments, resulting in contraction. Agency RMBS are not typically structured with tranches.

Exhaustion of a support tranche is a source of extension risk for a planned amortization class of a CMO.

For Further Reference:

(Study Session 14, Modules 45.1, 45.2, LOS 45.e)

CFA® Program Curriculum, Volume 5, page 194

Question #98 of 120

Question ID: 1207349

A hedge fund manager is estimating a value for a non-traded bond of Yoder Company. The bond has an annual-pay 6% coupon, matures in six years, and has a CCC credit rating. Actively traded annual-pay bonds with similar credit ratings include the following:

Coupon	Maturity	Yield to maturity
8%	5 years	9.45%
5%	5 years	9.55%
7%	10 years	10.00%

Based on matrix pricing, the value of the Yoder bond as a percentage of par is *closest* to:

- A) 83.9.
- B) **84.1.**
- C) 84.5.

Explanation

To find an appropriate discount rate for the Yoder bond, first take the average of the 5-year bonds:

$$(9.45\% + 9.55\%) / 2 = 9.50\%$$

Next, use linear interpolation to estimate a yield for a bond with six years to maturity:

$$9.50\% + [(6 - 5) / (10 - 5)] \times (10.00\% - 9.50\%) = 9.60\%$$

Finally, discount the Yoder bond's cash flows at this rate:

$$N = 6; I/Y = 9.6; PMT = 6; FV = 100; CPT PV = -84.14$$

For Further Reference:

(Study Session 14, Modules 44.1, 44.2, LOS 44.a, 44.e)

Question #99 of 120

Question ID: 1207357

Consider two option-free, 5% annual-pay bonds from the same issuer and with the same seniority. One of the bonds has a modified duration of 3.5 and approximate convexity of 15. The other has a modified duration of 8.5 and approximate convexity of 75. Can the lower- duration bond have more price volatility than the higher-duration bond?

- A) No, because it also exhibits lower convexity.
- B) Yes, because shifts in the yield curve may be non-parallel.**
- C) No, because its price will respond relatively less in response to changes in yield.

Explanation

Duration-based estimates of bond value changes assume the yield curve shifts in a parallel manner. If instead short-term interest rates are more volatile than long-term interest rates, it is possible for a bond with lower duration to have more price volatility than a bond with higher duration.

For Further Reference:

Study Session 17, Module 46.3, LOS 46.f

CFA® Program Curriculum, Volume 5, page 278

Question #100 of 120

Question ID: 1207368

Questions 100 through 106 relate to Derivatives. (10.5 minutes)

When the underlying asset does not pay any cash flows, the value of an American call option is:

- A) equal to the value of an otherwise identical European call option.**
- B) less than the value of an otherwise identical European call option.
- C) greater than the value of an otherwise identical European call option.

Explanation

Because the right to exercise a call option early is not valuable when the underlying asset does not pay any cash flows, the value of an American call option is equal to the value of an otherwise identical European call option.

For Further Reference:

(Study Session 16, Module 49.4, LOS 49.o)

CFA® Program Curriculum, Volume 5, page 500

Question #101 of 120

Question ID: 1207366

The value of a put option on a stock trading at 35 will decrease when:

- A) the risk-free rate is higher.**
- B) the volatility of the stock price is higher.
- C) the dividends paid on the stock are higher.

Explanation

An increase in the risk-free rate will decrease the value of a put option. An increase in the volatility of the stock price or a higher dividend payment during the option's life will increase the value of a put option.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.k)

CFA® Program Curriculum, Volume 5, page 482

Question #102 of 120

Question ID: 1207364

During the life of a European option, the amount by which its price is greater than its exercise value is *most accurately* described as its:

- A) time value.**
- B) moneyness.
- C) intrinsic value.

Explanation

Before expiration, an option can have a price greater than its exercise or intrinsic value. This amount by which an option's price is greater than its exercise value is referred to as its time value.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.j)

CFA® Program Curriculum, Volume 6, page 86

Question #103 of 120

Question ID: 1207362

Which of the following statements about plain vanilla interest rate swaps is *most accurate*? The swap counterparties:

- A) typically make a margin deposit with a clearinghouse.
- B) exchange fixed rate payments for variable rate payments.**
- C) exchange the notional principal at initiation and termination.

Explanation

The counterparties do not exchange the notional principal on an interest rate swap.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.c)

CFA® Program Curriculum, Volume 5, page 394

Question #104 of 120

Question ID: 1207363

The convenience yield associated with holding the underlying asset of a derivative is *most accurately* described as:

- A) the nonmonetary benefits of holding the asset.**
- B) the monetary and nonmonetary benefits of holding the asset.
- C) the monetary and nonmonetary benefits of holding the asset, net of its holding costs.

Explanation

Convenience yield refers to the nonmonetary benefits of holding an asset, for example being in a position to sell an overvalued asset that is difficult to sell short. Convenience yield does not include monetary benefits such as interest and dividend income. The costs of holding the asset, net of the monetary and nonmonetary benefits of holding it, is referred to as the net cost of carry.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.e)

CFA® Program Curriculum, Volume 5, page 471

Question #105 of 120

Question ID: 1207361

Which of the following statements about futures margin is *least accurate*?

- A) The initial margin is set by the clearinghouse based on the volatility of the price of the underlying asset.**
- B) If the balance of the margin account exceeds the initial margin requirement, the trader can remove the excess funds from the account.
- C) If the margin account balance falls below the maintenance margin level, the account balance must be brought back up to the maintenance level.**

Explanation

If the margin account balance falls below the maintenance margin level, the account must be brought back up to the initial margin amount.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.c)

CFA® Program Curriculum, Volume 5, page 394

Question #106 of 120

Question ID: 1207367

Which of the following portfolios has the same future cash flows as a put option?

- A) Long call option, long risk-free bond, short underlying asset.**
- B) Long call option, short risk-free bond, long underlying asset.
- C) Short call option, long risk-free bond, long underlying asset.

Explanation

Using the put-call parity relationship, a synthetic put option can be created by combining a long call option with the same exercise price and expiration date, a short position in the underlying asset, and a long position in a risk-free bond that pays the exercise price on the expiration date.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.I)

CFA® Program Curriculum, Volume 5, page 491

Question #107 of 120

Question ID: 1207372

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

If the availability of a physical commodity over the period of a futures contract has value to users of the commodity, the commodity is said to provide:

- A) storage yield.
- B) collateral yield.
- C) convenience yield.**

Explanation

Convenience yield is the value of a physical commodity being available for use over the period of a futures contract.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 145

Question #108 of 120

Question ID: 1207370

Josh Lacy, CFA, is analyzing a portfolio company held by his private equity firm to estimate its value in liquidation. Lacy should *most appropriately* use:

- A) an asset-based approach.**
- B) a comparables-based approach.
- C) a discounted cash flow-based approach.

Explanation

The asset-based approach uses either the liquidation values or fair market values of assets. The discounted cash flow approach involves calculating the present value of expected future cash flows. The comparables-based approach uses market or private transaction values of similar companies to estimate multiples of EBITDA, net income, or revenue.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 145

Question #109 of 120

Question ID: 1207373

Which of the following statements with respect to hedge fund investing is *least accurate*?

- A) Hedge funds only publicly disclose performance information on a voluntary basis.
- B) Hedge funds are not typically registered with the SEC in the United States.
- C) **Survivorship bias in hedge fund data causes risk to be overstated because funds that take on more risk tend to have higher returns.**

Explanation

Because a hedge fund database only includes the more stable funds that have survived, the risk measure of hedge funds as an asset class is biased downward.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 145

Question #110 of 120

Question ID: 1207374

An investor who wants to hedge against inflation by allocating a portion of a portfolio to alternative investments should *most appropriately* invest in:

- A) **real estate and commodities.**
- B) private equity and real estate.
- C) commodities and private equity.

Explanation

Real estate and commodities offer potential hedges against inflation because rents, property values, and commodity prices tend to increase with inflation in the long term.

For Further Reference:

(Study Session 17, Modules 50.1, 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #111 of 120

Question ID: 1207369

A leveraged buyout firm that carries out a secondary sale has:

- A) offered additional shares to the public.
- B) **exited an investment in a portfolio company.**
- C) received new capital from its general or limited partners.

Explanation

In a secondary sale, a private equity firm sells one of its portfolio companies to a group of investors or another private equity firm.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #112 of 120

Question ID: 1207375

The notice period for a hedge fund is *best* described as the period following:

- A) the opening of the fund to investors, before the fund is closed to new investors.
- B) a request for redemption of shares, within which the fund must fulfill the request.**
- C) an investment in the fund, during which the investor is not permitted to redeem shares.

Explanation

The notice period is the time within which a hedge fund must fulfill a request for redemption of shares. The period during which investors may not redeem shares is called a lockup period.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #113 of 120

Question ID: 1207371

Returns calculated from which type of real estate index are *most likely* to have the lowest standard deviation?

- A) REIT index.
- B) Appraisal index.**
- C) Repeat sales index.

Explanation

Valuations based on appraisals tend to smooth returns compared to using market-based valuations. As a result, returns based on an appraisal index are likely to have a lower standard deviation than returns based on a repeat sales index or a REIT index.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 145

Question #114 of 120

Question ID: 1207382

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

A financial firm employs machine learning to model its risk exposures. The machine identifies a number of risk relationships in the input data, but the firm's management believes some of these relationships are spurious. If so, it is *most likely* that the model:

- A) exhibits overfitting.**
- B) treats true parameters as noise.**
- C) is not complex enough to describe the data.**

Explanation

Spurious relationships and patterns are likely to result from overfitting, which describes a model that is too complex and treats noise as true parameters.

For Further Reference:

(Study Session 19, Module 57.1, LOS 57.b, 57.c)

Question #115 of 120

Question ID: 1207376

On average, which of the following types of investors has the shortest investment horizon?

- A) Endowment fund.**
- B) Defined benefit plan.**
- C) Property and casualty insurer.**

Explanation

Property and casualty insurers may need to distribute funds at any time if a disaster occurs. Endowment funds have a very long horizon, as they are often expected to operate in perpetuity with only annual distributions representing a relatively small percentage of the fund value. Defined benefit plans have an investment horizon based on the years to retirement of currently covered workers and the years they (and their beneficiaries) are expected to live after retirement.

For Further Reference:

(Study Session 18, Module 51.1, LOS 51.b)

CFA® Program Curriculum, Volume 6, page 89

Question #116 of 120

Question ID: 1207378

A portfolio is invested 30% in Asset A with the remainder invested in Asset B. Asset A has an expected return of 6% and variance of returns of 0.031, while Asset B has an expected return of 7% and variance of returns of 0.045. The covariance between the returns of the two assets is 0.03735. The standard deviation of returns for the portfolio is *closest* to:

- A) 18%.**
- B) 20%.**
- C) 22%.**

Explanation

The correlation between the returns of the two assets is:

$$\frac{\text{Cov}_{AB}}{\sigma_A \sigma_B} = \frac{0.03735}{\sqrt{0.031} \times \sqrt{0.045}} = 1$$

Therefore, the standard deviation of the portfolio returns is a weighted average of the standard deviations of returns for the two assets:
 $0.3\sqrt{0.031} + 0.7\sqrt{0.045} = 20.13\%$

Since the correlation of returns is +1, there are no diversification benefits.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.f)

CFA® Program Curriculum, Volume 6, page 150

Question #117 of 120

Question ID: 1207377

Open-end mutual funds differ from closed-end funds in that:

- A) open-end funds stand ready to redeem their shares, while closed-end funds do not.**
- B) closed-end funds require active management, while open-end funds do not.
- C) open-end funds issue shares that are then traded in secondary markets, while closed-end funds do not.

Explanation

Open-end funds redeem existing shares or issue new shares in accordance with investor demand. Closed-end fund shares are fixed in number and trade on exchanges as though they were common stock.

For Further Reference:

(Study Session 18, Module 51.2, LOS 51.f)

CFA® Program Curriculum, Volume 6, page 104

Question #118 of 120

Question ID: 1207379

According to the CAPM, a rational investor would be *least likely* to choose as his optimal portfolio:

- A) a 100% allocation to the risk-free asset.**
- B) the global minimum variance portfolio.
- C) a 130% allocation to the market portfolio.

Explanation

According to the CAPM, rational, risk-averse investors will optimally choose to hold a portfolio along the capital market line. This can range from a 100% allocation to the risk-free asset to a leveraged position in the market portfolio constructed by borrowing at the risk-free rate to invest more than 100% of the portfolio equity value in the market portfolio. The global minimum variance portfolio lies below the CML and is not an efficient portfolio under the assumptions of the CAPM.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.h)

CFA® Program Curriculum, Volume 6, page 163

Question #119 of 120

Question ID: 1207380

The risk-free rate is 5% and the expected market risk premium is 10%. A portfolio manager is projecting a return of 20% on a portfolio with a beta of 1.5. After adjusting for its systematic risk, this portfolio is expected to:

- A) equal the market's performance.**
- B) outperform the market.
- C) underperform the market.

Explanation

Based on the CAPM, the portfolio should earn: $E(R) = 0.05 + 1.5(0.10) = 20\%$. On a risk-adjusted basis, this portfolio lies on the security market line (SML) and thus is earning a risk-adjusted rate of return equivalent to that of the market portfolio.

For Further Reference:

(Study Session 18, Module 53.2, LOS 53.g)

CFA® Program Curriculum, Volume 6, page 213

Question #120 of 120

Question ID: 1207381

A manager states that the objectives of a firm's risk management process should be to:

- Identify the firm's risk tolerance.
- Identify and measure risks faced by the organization.
- Minimize or eliminate these risks.

Which of these objectives is *least appropriate* in a risk management process?

- A) Identifying risk tolerance.**
- B) Identifying and measuring risks.
- C) **Minimizing or eliminating risks.**

Explanation

Not all risks should be minimized or eliminated. A firm might choose to increase its exposure to some identified risks if it can profit from its ability to manage them.

For Further Reference:

(Study Session 19, Module 55.1, LOS 55.a)

CFA® Program Curriculum, Volume 6, page 292